Honours Project 2026: Tools and Code Used

A bioinformatics workflow of the tools used for the Honors project of Nicole Nakiyingi (25006096) at Stellenbosch University.

Software and tools used:

1. FastQC v0.12.1
2. MultiQC v1.35
3. Shovill v1.4.1
4. QUAST v5.3.0
5. Bakta v1.12.0
6. Panaroo v1.8.0
7. IQ-TREE v3.1.3
8. ABRicate v1.4.0

FastQC and MultiQC

```
conda activate fastqc

mkdir -p fastqc_output

fastqc *.gz -o fastqc_output

cd fastqc_output

multiqc .
```

Quality check of the trimmed sequenced datasets

Read assembly -Shovill

```
conda activate shovill

for f in *R1*.fastq.gz
do
    r2=${f/R1/R2}
    sample=${f%%_R1*}

    shovill \
        --R1 "$f" \
        --R2 "$r2" \
        --outdir "${sample}_Output" \
        --gsize 2.5M \
        --cpus 16 \
        --minlen 500 \
        --nocorr

done
```

The trimmed reads were assembled using Shovill, which integrates SPAdes. Post-assembly polishing was removed (using the --nocorr parameter) and done manually – minlen 500 (which also removed homopolymers).

Assembly quality assessment- QUAST

```
for sample in *VM110003* *VM110172* *VM110027*
do
    echo "Checking $sample"

    if [ -f "$sample/contigs.fa" ]
    then
        echo "Running QUAST on $sample"

        quast.py "$sample/contigs.fa" \
            -o "QC/quast/$sample"
    else
        echo "No contigs.fa found for $sample"
    fi

done
```

CheckM2

```
for sample in *VM110003* *VM110172* *VM110027*
do
    echo "Running CheckM2 on $sample"

    checkm2 predict \
        --threads 16 \
        --input "$sample/contigs.fa" \
        --output-directory "QC/checkM2/$sample"

done
```

#CheckM2 analyzed predicted completeness and contamination and helped remove 2 isolates from further analysis as they had high contamination scores.

Annotation using Bakta

```
conda activate bakta

mkdir -p bakta_results

for sample in \
    *VM110003*/contigs.fa \
    *VM110027*/contigs.fa \
    *VM110172*/contigs.fa
do
    name=$(basename "$(dirname "$sample")")

    bakta \
        --db /Applications/biotools/bakta_db/db/ \
        --output "bakta_results/$name" \
        "$sample"
```

```
done
```

Panaroo- for all *L. crispatus* isolates

#Panaroo was run on all L. crispatus isolates to build a full combined pangenome for pangenome coverage maximization.

```
panaroo \
    -i \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN13.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN07.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN12.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN08.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN14.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN15.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN01.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN09.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN11.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN05.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN10.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN16.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN06.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
003BKEN02.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
027*.gff3 \

~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/gff3/VM110
172*.gff3 \
```

```
    -o
~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/All_panaro
o \
    --clean-mode moderate \
    --remove-invalid-genes \
    -a core \
    --aligner mafft \
    --threads 4
```

Panaroo 2- for VM110027 isolates

#The individual women's pangenomes were also assembled to determine genetic diversity within women.

```
conda activate panaroo

panaroo \
    -i
~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Final_isolates/VM1
10027*/*.gff3 \
    -o
~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Final_isolates/VM1
10027_panaroo \
    --clean-mode moderate \
    --remove-invalid-genes \
    -a core \
    --aligner mafft \
    --threads 4
```

Panaroo 3- for VM110172 isolates

```
conda activate panaroo

panaroo \
    -i
~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Final_isolates/VM1
10172*/*.gff3 \
    -o
~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Final_isolates/VM1
10172_panaroo \
    --clean-mode moderate \
    --remove-invalid-genes \
    -a core \
    --aligner mafft \
    --threads 4
```

Panaroo 4- for VM110003 *L. crispatus* isolates

```
conda activate panaroo

panaroo \
    -i \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN13.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN07.gff3 \
```

```
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN12.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN08.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN14.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN15.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN01.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN09.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN11.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN05.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN10.gff3 \
    ~/Documents/brian/nicole_read_data_2/Panaroo/gff3/VM110003BKEN16.gff3 \
    -o ~/Documents/brian/nicole_read_data_2/Panaroo/VM110003_Part_1_panaroo
\
    --clean-mode moderate \
    --remove-invalid-genes \
    -a core \
    --aligner mafft \
    --threads 4
```

IQ-TREE 3- for all *L. crispatus* isolates

#Phylogenetic trees were produced using IQtree and visualized using ITOL, where the different groupings were observed.

```
iqtree3 \
    -s
~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/All_2_pana
roo/core_gene_alignment_filtered.aln \
    -p
~/Documents/brian/nicole_read_data_2/Partition_files/core_alignment_filtere
d_header_All.partition \
    -m MFP+MERGE \
    -B 1000 \
    -T AUTO \
    --prefix ~/Documents/brian/nicole_read_data_2/IQtree/All_IQtree
```

IQ-TREE 3- for VM110003 *L. crispatus* isolates

```
iqtree3 \
    -s
~/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/VM110003_P
art_5_LC_panaroo/core_gene_alignment_filtered.aln \
    -p /Users/ceri-mac-
1/Documents/brian/nicole_read_data_2/Partition_files/core_alignment_filtere
d_header_VM110003_LC.partition \
    -m MFP+MERGE \
    -B 1000 \
    -T AUTO \
    --prefix /Users/ceri-mac-
1/Documents/brian/nicole_read_data_2/IQtree/VM110003_LC_IQtree
```

IQ-TREE 3- for VM110027 isolates

```
iqtree3 \
    -s /Users/ceri-mac-
1/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/VM110027_p
anaroo/core_gene_alignment_filtered.aln \
    -p /Users/ceri-mac-
1/Documents/brian/nicole_read_data_2/Partition_files/core_alignment_filtere
d_header_VM110027.partition \
    -m MFP+MERGE \
    -B 1000 \
    -T AUTO \
    --prefix /Users/ceri-mac-
1/Documents/brian/nicole_read_data_2/IQtree/VM110027_IQtree
```

IQ-TREE 3- for VM110172 isolates

```
iqtree3 \
    -s /Users/ceri-mac-
1/Documents/brian/nicole_read_data_2/Bakta_annotations_2/Panaroo/VM110172_p
anaroo/core_gene_alignment_filtered.aln \
    -p /Users/ceri-mac-
1/Documents/brian/nicole_read_data_2/Partition_files/core_alignment_filtere
d_header_VM110172.partition \
    -m MFP+MERGE \
    -B 1000 \
    -T AUTO \
    --prefix /Users/ceri-mac-
1/Documents/brian/nicole_read_data_2/IQtree/VM110172_IQtree
```

ABRicate

```
conda activate abricate

mkdir -p card_results

for sample in \
    *VM110003*/contigs.fa \
    *VM110027*/contigs.fa \
    *VM110172*/contigs.fa
do
    name=$(basename "$(dirname "$sample")")

    abricate --db card "$sample" \
        > "card_results/${name}.txt"

done
```

Code for pangenome coverage maximization

L. crispatus Maximum Gene Coverage Analysis

#

Input:

gene_presence_absence_roary_All.csv.xlsx

```
#
# Search strategy:
#   k = 1–5  : exact exhaustive search
#   k >= 6   : greedy extension from the optimal k = 5 solution
#
# Note:
#   Results for k <= 5 are true maxima.
#   Results for k >= 6 are greedy approximations.
# Load required packages
library(readxl)
library(ggplot2)

# Select the input Excel file
input_file <- file.choose()

# Read the Roary output file
roary <- read_excel(
  input_file,
  .name_repair = "minimal"
)

# Identify isolate columns
isolate_columns <- 15:ncol(roary)
isolate_names <- colnames(roary)[isolate_columns]
n_isolates <- length(isolate_names)

# Initialise the gene presence/absence matrix
pa <- matrix(
```

```
  0L,
  nrow = nrow(roary),
  ncol = n_isolates
)

# Assign isolate names to matrix columns
colnames(pa) <- isolate_names

# Convert gene presence/absence data to binary values
for (j in seq_along(isolate_names)) {
  x <- as.character(roary[[isolate_names[j]]])

  pa[, j] <- as.integer(
    !is.na(x) & nzchar(trimws(x))
  )
}

# Ensure the matrix contains integer values
storage.mode(pa) <- "integer"

# Check for missing values
if (anyNA(pa)) {
  stop("Binary matrix contains NA values.")
}

# Calculate gene frequency across all isolates
gene_frequency_all <- rowSums(pa)
```

```
# Identify genes present in at least one isolate
keep_genes <- gene_frequency_all > 0

# Identify rows with no gene presence
blank_genes <- gene_frequency_all == 0

# Summarise the number of rows
total_rows_excel <- nrow(roary)
blank_gene_rows <- sum(blank_genes)
```

Code for Pairwise and phylogenetic heatmap